# A Scalable Interior-Point Gauss-Newton Method for PDE-Constrained Optimization with Bound Constraints

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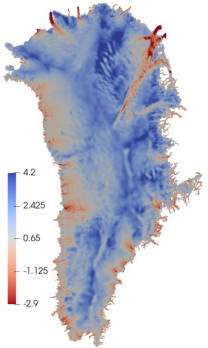
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# Motivation – PDE-constrained optimization with bound constraints

- **Example:** inversion (from surface flow observations) for the *non-negative* ice sheet basal friction field.
- **Challenge:** inequality constraints lead to computational challenges via complementarity in the KKT conditions.
- **Goal:** develop a scalable computational framework for such PDE- and bound-constrained optimization problems.




Picture from: T. Hartland, G. Stadler, K. Liegeois, M. Perego, and N. Petra. "Hierarchical off-diagonal low-rank approximation of Hessians in inverse problems, with application to ice sheet model initialization", Inverse Problems, 2023.

# PDE- and bound-constrained optimization problem structure

## General problem statement

$$\min_{(u,\rho)} \mathcal{J}(u,\rho) := \mathcal{J}_{\text{misfit}}(u) + \mathcal{J}_{\text{reg}}(\rho)$$

such that

$$\rho(t,\boldsymbol{x}) \geq \rho_\ell(t,\boldsymbol{x}), \text{ on } [0,T] \times \overline{\Omega} \text{ and}$$
$$r(u,\rho) = 0, \text{ in } (0,T) \times \Omega$$

**Notation:**

1. $u = u(t,\boldsymbol{x}), \rho = \rho(t,\boldsymbol{x})$, the state and parameter (respectively).
2. $\Omega$ – spatial domain, $[0,T]$ – time domain.
3. $\rho_\ell$ – lower-bound constraint, $r$ – PDE residual.

# PDE- and bound-constrained optimization common approach

General problem **common "reduced-space" approach**

$$\min_{\beta} \mathcal{J}(\beta) := \mathcal{J}(u(\rho), \rho), \quad \rho = \exp(\beta) + \rho_\ell$$

where $u(\rho)$ defined implicitly by $r(u(\rho), \rho) = 0$

Pros and cons

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+ The optimization problem is a reduced unconstrained problem and so simple optimization methods (Newton, Gauss-Newton, etc) are applicable.

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Pros and cons

+ The optimization problem is a reduced unconstrained problem and so simple optimization methods (Newton, Gauss-Newton, etc) are applicable.
- Each objective evaluation requires a PDE-solve.
- The reparametrization $\rho = \exp(\beta) + \rho_\ell$ can introduce higher-order nonlinearities and does not make use of constrained numerical optimization research.

# PDE- and bound-constrained optimization example problem I

## Nonlinear elliptic example problem

$$\min_{(u,\rho)} \mathcal{J}(u,\rho) := \underbrace{\frac{1}{2}\int_{\Omega_{\text{obs}}} (u(\boldsymbol{x}) - u_d(\boldsymbol{x}))^2 \,\mathrm{d}\boldsymbol{x}}_{\text{data-misfit}} + \underbrace{\frac{1}{2}\int_{\Omega} (\gamma_1 \rho(\boldsymbol{x})^2 + \gamma_2 \boldsymbol{\nabla}\rho \cdot \boldsymbol{\nabla}\rho)\mathrm{d}\boldsymbol{x}}_{\text{regularization}}$$

such that

$$\rho(\boldsymbol{x}) \geq \rho_\ell(\boldsymbol{x}), \text{ on } \overline{\Omega} = [0,1]^2 \text{ and}$$

$$\begin{cases} -\boldsymbol{\nabla}\cdot(\rho\,\boldsymbol{\nabla} u) + u + u^3/3 &= g \quad \text{in } \Omega \\ \rho\,\boldsymbol{\nabla} u \cdot \boldsymbol{n} &= 0 \quad \text{on } \partial\Omega \end{cases}$$

# PDE- and bound-constrained optimization example problem II

## Linear parabolic example problem

$$\min_{(u,\rho)} \mathcal{J}(u,\rho) := \underbrace{\frac{1}{2}\int_{\Omega_{\text{obs}}} (u(T,\boldsymbol{x}) - u_d(\boldsymbol{x}))^2 \,\mathrm{d}\boldsymbol{x}}_{\text{data-misfit}} + \underbrace{\frac{1}{2}\int_{\Omega} (\gamma_1 \rho(\boldsymbol{x})^2 + \gamma_2 \boldsymbol{\nabla}\rho \cdot \boldsymbol{\nabla}\rho)\mathrm{d}\boldsymbol{x}}_{\text{regularization}}$$

such that

$\rho(\boldsymbol{x}) \geq \rho_\ell(\boldsymbol{x})$, on $\overline{\Omega} = [0,1]^2$ (periodic) and

$$\begin{cases} \partial u/\partial t - \boldsymbol{\nabla}\cdot(\kappa\,\boldsymbol{\nabla} u) & = g(\boldsymbol{x}) \quad \text{in } (0,T)\times\overline{\Omega} \\ u(t,\boldsymbol{x})|_{t=0} & = \rho(\boldsymbol{x}) \quad \text{in } \overline{\Omega} \end{cases}$$

# Infinite-dimensional interior-point (IP) optimality system

**1.** Cast the PDE into weak form (assumed here time independent)

$$\text{Find } u \in H^1(\Omega) \text{ such that} \quad c(u, \rho, \lambda) = 0, \ \ \forall \lambda \in H^1(\Omega).$$

**2.** Define the log-barrier penalized Lagrangian function with PDE-constraint Lagrange multiplier $\lambda$ and log-barrier penalty parameter $\mu > 0$

$$\mathcal{L}(u, \rho, \lambda) := \underbrace{\mathcal{J}(u, \rho)}_{\text{Objective}} \underbrace{- \mu \int_\Omega \log(\rho - \rho_\ell) \, \mathrm{d}\boldsymbol{x}}_{\text{log-barrier term}} + \underbrace{c(u, \rho, \lambda)}_{\text{PDE constraint}} .$$

**3.** Solve the interior-point regularized nonlinear optimality system for $\mu \to 0^+$

$$\mathcal{L}_u \tilde{u} = 0, \ \ \forall \tilde{u} \in H^1(\Omega), \qquad \{\text{stationarity}\}$$

$$\mathcal{L}_\rho \tilde{\rho} = 0, \ \ \forall \tilde{\rho} \in H^1(\Omega) \cap L^\infty(\Omega), \qquad \{\text{stationarity}\}$$

$$\mathcal{L}_\lambda \tilde{\lambda} = 0, \ \ \forall \tilde{\lambda} \in H^1(\Omega), \qquad \{\text{feasibility}\}$$

# "Outer" IP-Gauss-Newton method

**1.** Construct the $\mu > 0$, nonlinear continuation system

$$\mathcal{L}_u \tilde{u} = 0, \quad \forall \tilde{u} \in H^1(\Omega), \qquad \{\text{stationarity}\}$$
$$\mathcal{L}_\rho \tilde{\rho} = 0, \quad \forall \tilde{\rho} \in H^1(\Omega) \cap L^\infty(\Omega), \qquad \{\text{stationarity}\}$$
$$\mathcal{L}_\lambda \tilde{\lambda} = 0, \quad \forall \tilde{\lambda} \in H^1(\Omega), \qquad \{\text{feasibility}\}$$

**2.** Use the Gauss-Newton method, with stopping criteria defined by mass-weighted norms, to inexactly solve log-barrier subproblems ($\mu \searrow 0$), with a filter line-search IPM for robust convergence

More details can be found in:


A. Wächter and L.T. Biegler. *On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming*, Mathematical Programming, 2006.

T. Hartland, C.G. Petra, N. Petra, J. Wang. *A scalable interior-point Gauss-Newton method for PDE-constrained optimization with bound constraints*, arXiv, 2024 (in review).

# IP-Gauss-Newton linear solve is a critical computational step

– The IP-Gauss-Newton linear system

$$\underbrace{\begin{pmatrix} \boldsymbol{H}_{u,u} & \boldsymbol{0} & \boldsymbol{J}_u^\top \\ \boldsymbol{0} & \boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}} & \boldsymbol{J}_\rho^\top \\ \boldsymbol{J}_u & \boldsymbol{J}_\rho & \boldsymbol{0} \end{pmatrix}}_{\boldsymbol{A}} \begin{pmatrix} \hat{\boldsymbol{u}} \\ \hat{\boldsymbol{\rho}} \\ \hat{\boldsymbol{\lambda}} \end{pmatrix} = \begin{pmatrix} \boldsymbol{b}_u \\ \boldsymbol{b}_\rho \\ \boldsymbol{b}_\lambda \end{pmatrix}$$

that must be solved for the search direction $(\hat{\boldsymbol{u}}, \hat{\boldsymbol{\rho}}, \hat{\boldsymbol{\lambda}})$ at each "outer" optimization step.

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– $\boldsymbol{J}_{\boldsymbol{u}}$, discretized linearized forward PDE, $\boldsymbol{R} = \gamma_1 \boldsymbol{M} + \gamma_2 \boldsymbol{K}$, regularization

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– $\boldsymbol{J}_u$, discretized linearized forward PDE, $\boldsymbol{R} = \gamma_1 \boldsymbol{M} + \gamma_2 \boldsymbol{K}$, regularization

– $\boldsymbol{H}_{\text{log-barrier}} = \boldsymbol{M}_{\text{lumped}} \boldsymbol{D}$, the log-barrier Hessian, a **positive** definite **diagonal** matrix that generally is ***very*** **ill-conditioned** ($\mu \searrow 0$).

# IP-Gauss-Newton linear system preconditioner

- **Goal:** efficient iterative solution of the IP-Gauss-Newton linear system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$
- **Strategy 1:** GMRES with the block Gauss-Seidel preconditioner,

$$\tilde{\boldsymbol{A}} := \begin{pmatrix} \boldsymbol{H}_{u,u} & \boldsymbol{0} & \boldsymbol{J}_u{}^\top \\ \boldsymbol{0} & \boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}} & \boldsymbol{J}_\rho{}^\top \\ \boldsymbol{J}_u & \boldsymbol{0} & \boldsymbol{0} \end{pmatrix}.$$


J. Pestana, T. Rees. *Null-space preconditioners for saddle point systems*, SIAM Journal on Matrix Analysis and Applications, 2016

G. Biros, O. Ghattas. *Parallel Lagrange-Newton-Krylov-Schur methods for PDE-constrained optimization. Part I: The Krylov-Schur solver*, SIAM Journal on Scientific Computing, 2005


- **Strategy 2:** Log-barrier and regularization, $\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}}$ preconditioned CG for the equivalent "reduced-space" Schur complement system

# The Gauss-Seidel preconditioned IP-Gauss-Newton matrix

The preconditioned matrix $\tilde{\boldsymbol{A}}^{-1}\boldsymbol{A}$, is similar to $\begin{bmatrix} \boldsymbol{I} & \begin{bmatrix} \boldsymbol{H}_{u,u} & \boldsymbol{J}_{\boldsymbol{u}}^{\top} \\ \boldsymbol{J}_{\boldsymbol{u}} & \boldsymbol{0} \end{bmatrix}^{-1} \begin{bmatrix} \boldsymbol{0} & \boldsymbol{J}_{\rho}^{\top} \end{bmatrix} \\ \boldsymbol{0} & \boldsymbol{I} + (\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}})^{-1}\hat{\boldsymbol{H}}_{\boldsymbol{d}} \end{bmatrix}$.

$\hat{\boldsymbol{H}}_{\boldsymbol{d}}$ is the positive semi-definite "reduced-space" data-misfit Gauss-Newton Hessian

$$\hat{\boldsymbol{H}}_{\boldsymbol{d}} = (\boldsymbol{J}_{\boldsymbol{u}}{}^{-1}\boldsymbol{J}_{\rho})^{\top}\boldsymbol{H}_{u,u}(\boldsymbol{J}_{\boldsymbol{u}}{}^{-1}\boldsymbol{J}_{\rho})$$

Particular to typical ill-posed PDE-constrained optimization problems:

- eigenvalues of $\boldsymbol{R}^{-1}\hat{\boldsymbol{H}}_{\boldsymbol{d}}$ decay rapidly to zero and in a mesh independent fashion

More details can be found in:

O. Ghattas, K. Wilcox. *Learning physics-based models from data: perspectives from inverse problems and model reduction*, Acta Numerica (2021).

# Block Gauss-Seidel preconditioner effectively clusters spectrum

Bounds on the eigenvalues of preconditioned matrix

$$1 \leq \lambda_j(\tilde{\boldsymbol{A}}^{-1}\boldsymbol{A}) \leq \begin{cases} 1 + \lambda_j(\boldsymbol{R}^{-1}\hat{\boldsymbol{H}}_{\boldsymbol{d}}), & 1 \leq j \leq \dim(\boldsymbol{\rho}), \\ 1, & \dim(\boldsymbol{\rho}) + 1 \leq j \leq \dim(\boldsymbol{\rho}) + 2 \cdot \dim(\boldsymbol{u}), \end{cases}$$

- eigenvalues of $\boldsymbol{R}^{-1}\hat{\boldsymbol{H}}_{\boldsymbol{d}}$ decay rapidly to zero and in a mesh independent fashion
- $\boldsymbol{R}^{-1}\hat{\boldsymbol{H}}_{\boldsymbol{d}}$ does not contain components specific to IPM

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Conclusions to be drawn from the eigenvalue bounds

– eigenvalues of $\tilde{\boldsymbol{A}}^{-1}\boldsymbol{A}$ decay rapidly to one and in a mesh independent fashion

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Conclusions to be drawn from the eigenvalue bounds

- eigenvalues of $\tilde{\boldsymbol{A}}^{-1}\boldsymbol{A}$ decay rapidly to one and in a mesh independent fashion
- spectrum of $\tilde{\boldsymbol{A}}^{-1}\boldsymbol{A}$ is largely independent of IPM ill-conditioning

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**Ill-conditioning due to IPM is factored out by the preconditioner.**

Details in: T. Hartland, C.G. Petra, N. Petra, J. Wang. *A scalable interior-point Gauss-Newton method for PDE-constrained optimization with bound constraints*, arXiv, 2024 (in review).

# The cost to apply the block Gauss-Seidel preconditioner

To compute

$$\boldsymbol{x} = \tilde{\boldsymbol{A}}^{-1}\boldsymbol{b} = \begin{pmatrix} \boldsymbol{H}_{u,u} & \boldsymbol{0} & \boldsymbol{J}_u{}^\top \\ \boldsymbol{0} & \boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}} & \boldsymbol{J}_\rho{}^\top \\ \boldsymbol{J}_u & \boldsymbol{0} & \boldsymbol{0} \end{pmatrix}^{-1} \begin{pmatrix} \boldsymbol{b}_u \\ \boldsymbol{b}_\rho \\ \boldsymbol{b}_\lambda \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}_u \\ \boldsymbol{x}_\rho \\ \boldsymbol{x}_\lambda \end{pmatrix}$$

1. $\boldsymbol{x}_u = \boldsymbol{J}_u{}^{-1}\boldsymbol{b}_\lambda$ {linearized forward PDE solve}
2. $\boldsymbol{x}_\lambda = \boldsymbol{J}_u{}^{-\top}\left(\boldsymbol{b}_u - \boldsymbol{H}_{u,u}\boldsymbol{x}_u\right)$ {adjoint PDE solve}
3. $\boldsymbol{x}_\rho = \left(\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}}\right)^{-1}\left(\boldsymbol{b}_\rho - \boldsymbol{J}_\rho{}^\top \boldsymbol{x}_\lambda\right)$ {AMG-CG solve}

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3. $\boldsymbol{x}_\rho = \left(\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}}\right)^{-1}\left(\boldsymbol{b}_\rho - \boldsymbol{J}_\rho{}^\top \boldsymbol{x}_\lambda\right)$ {AMG-CG solve}

The log-barrier Hessian improves the **diagonal dominance** of $\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}}$

$$\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}} = \gamma_1 \boldsymbol{M} + \gamma_2 \boldsymbol{K} + \underbrace{\boldsymbol{M}_{\text{lumped}}\boldsymbol{D}}_{\text{log-barrier Hessian}}$$

# Sketch of Preconditioned CG approach

Idea: form Schur complement system by eliminating $\hat{\boldsymbol{u}}$, $\hat{\boldsymbol{\lambda}}$

$$(\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}} + \hat{\boldsymbol{H}}_{\boldsymbol{d}})\hat{\boldsymbol{\rho}} = \boldsymbol{b}$$

Use preconditioner: $\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}}$. The eigenvalues

$$(\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}})^{-1} \left(\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}} + \hat{\boldsymbol{H}}_{\boldsymbol{d}}\right) = \boldsymbol{I} + (\boldsymbol{R} + \boldsymbol{H}_{\text{log-barrier}})^{-1} \hat{\boldsymbol{H}}_{\boldsymbol{d}}$$

of this preconditioned system are intimately related to the eigenvalues of $\tilde{\boldsymbol{A}}^{-1}\boldsymbol{A}$.

# MFEM-based implementation of the IP-Gauss-Newton framework

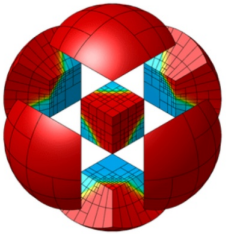
## Features utilized

- Modular object-oriented design;
- Distributed memory parallelism (MPI);
- Finite element discretization, mesh refinement, Krylov subspace solvers (MFEM);
- Algebraic multigrid preconditioners (hypre);

More details can be found in:

J. Andrej, et al. *High performance finite elements with MFEM*, The International Journal of High Performance Computing Applications, 2024.

hypre. *High Performance Preconditioners*, https://llnl.gov/casc/hypre



MFEM is a *free, lightweight, scalable* C++ library for finite element methods.



# Nonlinear elliptic PDE- and bound-constrained example problem

$$\min_{(u,\rho)} \mathcal{J}(u,\rho) := \underbrace{\frac{1}{2}\int_{\Omega_{\text{obs}}} (u(\boldsymbol{x}) - u_d(\boldsymbol{x}))^2 \,\mathrm{d}\boldsymbol{x}}_{\text{data-misfit}} + \underbrace{\frac{1}{2}\int_{\Omega} (\gamma_1 \rho(\boldsymbol{x})^2 + \gamma_2 \boldsymbol{\nabla}\rho \cdot \boldsymbol{\nabla}\rho)\mathrm{d}\boldsymbol{x}}_{\text{regularization}}$$

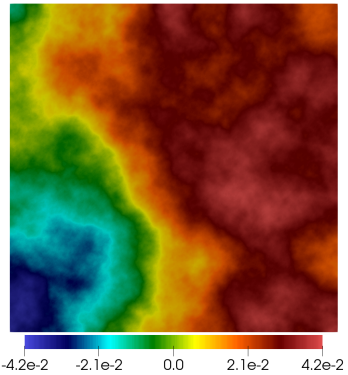
such that

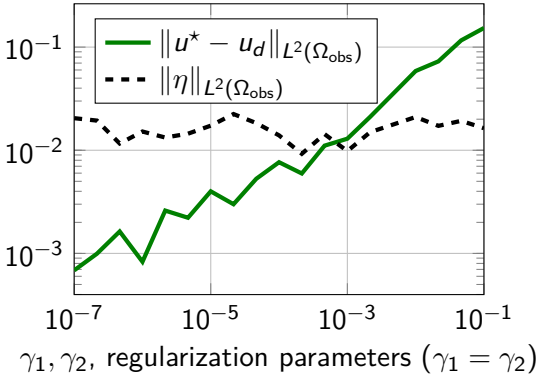
$$\rho(\boldsymbol{x}) \geq \rho_\ell(\boldsymbol{x}) = 1.0, \text{ on } \overline{\Omega} = [0,1]^2 \text{ and}$$

$$\begin{cases} -\boldsymbol{\nabla}\cdot(\rho\,\boldsymbol{\nabla} u) + u + u^3/3 & = g \quad \text{in } \Omega \\ \rho\,\boldsymbol{\nabla} u \cdot \boldsymbol{n} & = 0 \quad \text{on } \partial\Omega \end{cases}$$

- $\Omega_{\text{obs}} = (0, 1/2) \times (0, 1)$, observation domain
- $u_d(\boldsymbol{x}) = \cos(\pi\,\boldsymbol{x}_1)\cos(\pi\,\boldsymbol{x}_2) + \eta(\boldsymbol{x})$, noisy ($\eta$) data

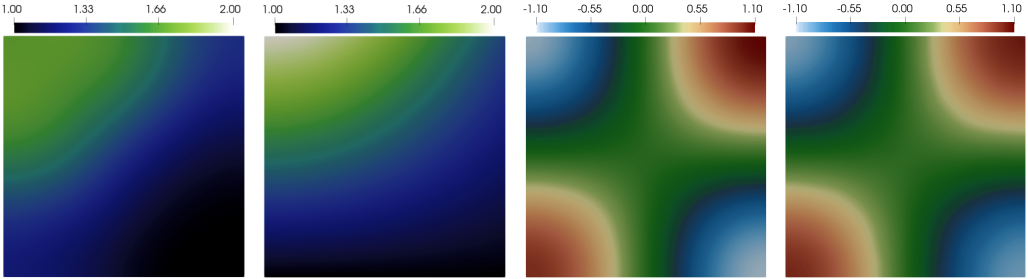
# Regularization determined by the Morozov discrepancy principle




Left: spatial structure of a random noise sample $\eta$ (5% relative noise level).
Right: Seminorms of the discrepancy $(u^\star - u_d)$ and noise $\eta$ as functions of the regularization.

# Nonlinear elliptic PDE- and bound-constrained problem solution




(a) $\rho^\star$ (b) $\rho_{\text{true}}$ (c) $u^\star$ (d) $u_d$

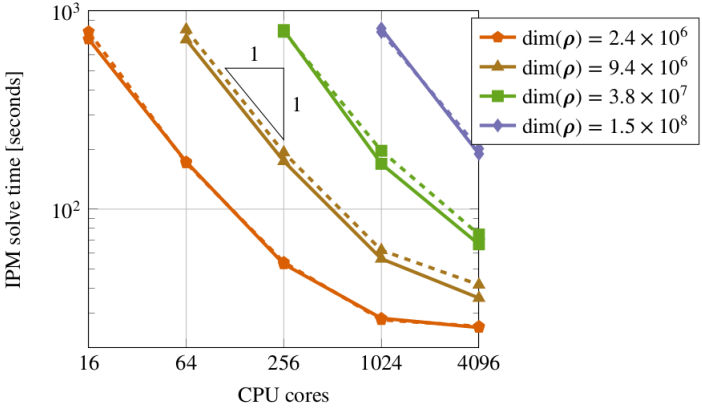
Computed optimum $\rho^\star$ (left), $\rho_{\text{true}}$ (middle left), computed optimum state $u^\star$ (middle right) and noisy state data $u_d$ (right).

# Solution computed with a mesh independent number of iterations

| $\dim(\rho)$ | Average interior-point per optimizer solve | Average GMRES iterations per linear solve | Average CG iterations per linear solve |
|---|---|---|---|
| 148 609 | 28.4 | 6.50 | 6.76 |
| 591 361 | 28.2 | 6.48 | 6.72 |
| 2 362 369 | 28.8 | 6.51 | 6.68 |
| 9 443 329 | 28.3 | 6.49 | 6.85 |
| 37 761 025 | 28.7 | 6.42 | 6.87 |
| 151 019 521 | 29.0 | 6.52 | 6.75 |

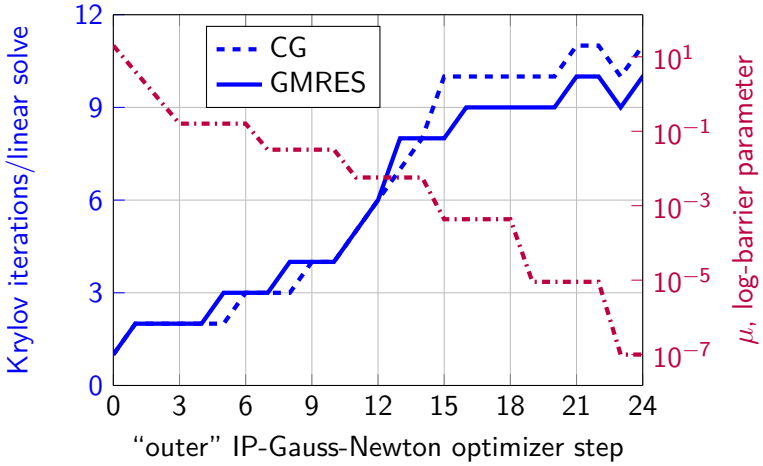
Figure: Outer interior-point optimizer and inner Krylov iteration counts. $10^{-6}$ interior-point optimizer absolute tolerance and $10^{-8}$ relative linear solve tolerance.

# Strong scaling (quartz) of the IP-Gauss-Newton-Krylov method




GMRES: solid lines, CG: dashed lines. Timings obtained on Intel Xeon E5-2695 v4 chips.

# Preconditioners are robust with respect to IPM ill-conditioning



# Parabolic PDE- and bound-constrained example problem

## Linear parabolic example problem

$$\min_{(u,\rho)} \mathcal{J}(u,\rho) := \underbrace{\frac{1}{2}\int_{\Omega_{\text{obs}}} (u(T,\boldsymbol{x}) - u_d(\boldsymbol{x}))^2 \,\mathrm{d}\boldsymbol{x}}_{\text{data-misfit}} + \underbrace{\frac{1}{2}\int_{\Omega} (\gamma_1 \rho(\boldsymbol{x})^2 + \gamma_2 \boldsymbol{\nabla}\rho \cdot \boldsymbol{\nabla}\rho)\mathrm{d}\boldsymbol{x}}_{\text{regularization}}$$

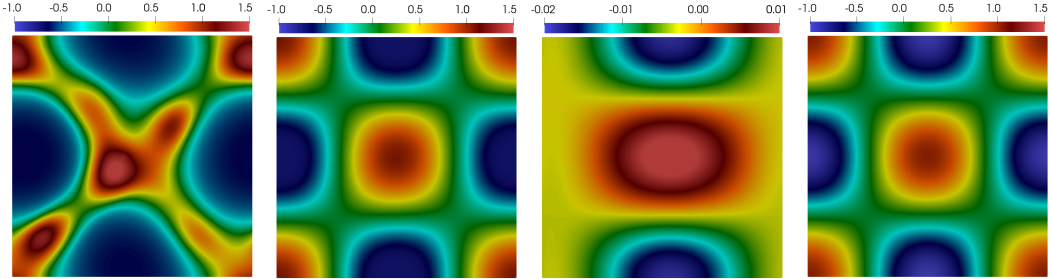
such that

$$\rho(\boldsymbol{x}) \geq \rho_\ell(\boldsymbol{x}) = -3/4, \text{ on } \Omega = (0,1)^2 \text{ (periodic) and}$$

$$\begin{cases} \partial u/\partial t - \boldsymbol{\nabla}\cdot(\kappa\,\boldsymbol{\nabla} u) & = g(\boldsymbol{x}) \quad \text{in } (0,T)\times\Omega \\ u(t,\boldsymbol{x})|_{t=0} & = \rho(\boldsymbol{x}) \quad \text{on } \Omega \end{cases}$$

- $\rho_{\text{true}}(\boldsymbol{x}) = \cos(\pi\,\boldsymbol{x}_1)\cos(\pi\,\boldsymbol{x}_2)$,

# Parabolic PDE- and bound-constrained problem solution




(a) $\rho^\star$ ($\gamma_2 = 10^{-9}$) (b) $\rho^\star$ ($\gamma_2 = 10^{-5}$) (c) $\rho^\star$ ($\gamma_2 = 10^{-1}$) (d) $\rho_{\text{true}}$

Parameter reconstruction $\rho^\star$ with various regularization parameters (left, middle left and middle right), and true parameter $\rho_{\text{true}}$ (right).

# Solution computed with a mesh independent number of iterations for a wide range of regularization parameter values

| $\dim(\boldsymbol{\rho})$ | # iter GMRES, (# iter IP) | | | | |
|---|---|---|---|---|---|
| | $\gamma_2 = 10^{-10}$ | $\gamma_2 = 10^{-8}$ | $\gamma_2 = 10^{-6}$ | $\gamma_2 = 10^{-4}$ | $\gamma_2 = 10^{-2}$ |
| $9.2 \times 10^3$ | 16.8 (14) | 16.7 (14) | 15.4 (13) | 11.8 (11) | 7.1 (7) |
| $3.7 \times 10^4$ | 18.7 (15) | 17.2 (14) | 15.4 (13) | 12.1 (11) | 7.4 (7) |
| $1.5 \times 10^5$ | 19.5 (15) | 18.9 (14) | 15.5 (13) | 11.9 (11) | 7.3 (7) |
| $5.9 \times 10^5$ | 20.2 (15) | 20.5 (15) | 15.4 (13) | 12.1 (11) | 7.1 (7) |
| $2.4 \times 10^6$ | 20.7 (14) | 20.1 (14) | 15.8 (13) | 11.9 (11) | 7.4 (7) |

Table: Algorithmic scaling of the IP-Gauss-Newton method with block Gauss-Seidel preconditioned GMRES solves for the parabolic time-dependent PDE- and bound-constrained optimization with backward Euler time step $\Delta t = 0.01$. The absolute tolerance of the outer optimization loop is $10^{-6}$ and the relative tolerance of the block AMG-CG solves is $10^{-13}$.

# Conclusions and future work

- Algorithmically scalable IP-Gauss-Newton method for PDE- and bound-constrained optimization problems that respects the nature of the infinite-dimensional problem.
- Ill-conditioning of IP-Gauss-Newton linear systems handled by preconditioners that exploit PDE-constrained optimization problem structure.
- Interested in applying this framework to a broader set of problems and discussions/feedback from the MFEM community.
- The IPM solver (without PDE-constrained examples) is available at https://github.com/LLNL/ContinuationSolvers.
- Details in: T. Hartland, C.G. Petra, N. Petra, J. Wang. *A scalable interior-point Gauss-Newton method for PDE-constrained optimization with bound constraints*, arXiv, 2024 (in review).

Thank you for your attention.

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